

Learning Resources — Where to Study Each of the 25 Topics

Primary = watch this first. Backup = use if primary does not click. Striver: best overall. Aditya Verma: best for Backtracking/Sliding Window/DP/Heaps. CP-Algorithms: best for Phase 5.

TOPIC	PRIMARY — watch this first	BACKUP — if primary does not click
Phase 1 — Foundations		
01. Complexity Analysis	Striver — Time & Space Complexity (YouTube, ~45 min). Search: 'Striver time complexity'	Abdul Bari — Asymptotic Notations (YouTube). More mathematical, good second watch.
02. C++ STL Essentials	Striver — C++ STL Full Series (YouTube, ~3 hrs). Search: 'Striver C++ STL'	CodeWithHarry — C++ STL Tutorial (YouTube). Slower pace for absolute beginners.
03. Recursion	Apna College — Recursion playlist, first 5 videos (YouTube). Search: 'Apna College recursion'	Striver — Recursion Series (YouTube). More problems, pattern-based approach.
04. Arrays	Striver — A-Z Sheet Arrays section Easy→Medium→Hard (YouTube + takeuforward.org)	Apna College — Arrays playlist (YouTube). Gentler pace, more intuition building.
05. Strings	Apna College — Strings playlist (YouTube). Search: 'Apna College strings'	Striver — A-Z Sheet Strings section (YouTube). Problem-first approach.
06. Linked List	Apna College — Linked List full playlist (YouTube). Best beginner LL series available.	Striver — Linked List Series (YouTube). More interview-focused problem set.
07. Stack & Queue	Striver — Stack & Queue Series (YouTube). Covers all patterns including monotonic stack.	Apna College — Stack & Queue playlist (YouTube). Good for conceptual understanding.
08. Hashing & Hashmaps	Striver — Hashing Series (YouTube). All hashmap patterns covered in ~3 videos.	NeetCode — Arrays & Hashing section (neetcode.io). Short tight explanations.
09. Searching	Striver — Binary Search Series (YouTube, ~25 videos). Most complete free BS resource.	Apna College — Binary Search playlist (YouTube). Slower pace, beginner friendly.
10. Sorting Algorithms	Apna College — Sorting playlist (YouTube). Implements all algorithms from scratch.	Striver — Sorting videos in A-Z Sheet (YouTube). More interview context.
11. Bit Manipulation	NeetCode — Bit Manipulation playlist (YouTube, 6 videos, one sitting ~1.5 hrs).	Striver — Bit Manipulation Series (YouTube). More problems with detailed walkthroughs.
Phase 2 — Patterns		
12. Two Pointers	Striver — Two Pointer Series (YouTube). Covers opposite and same direction variants.	NeetCode — Two Pointers section (neetcode.io). Concise pattern-focused explanations.
13. Sliding Window	Striver — Sliding Window Series (YouTube). Fixed and variable window patterns.	Aditya Verma — Sliding Window playlist (YouTube). Best beginner SW series, very patient.
14. Binary Trees	Apna College — Binary Trees full playlist (YouTube). Best beginner tree series.	Striver — Binary Tree Series (YouTube, ~50 videos). Very detailed, interview-focused.

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15. Binary Search Tree	Striver — BST Series (YouTube). All BST operations and interview problems covered.	Apna College — BST playlist (YouTube). More conceptual, good for invariant understanding.
16. Heaps & Priority Queue	Striver — Heaps Series (YouTube). Heap internals and all top-K patterns.	Aditya Verma — Heap playlist (YouTube). Pattern-based, very clear for beginners.
17. Math & Number Theory	NeetCode — Math section (neetcode.io) + GFG 'Modular Arithmetic' article (must read).	Striver — Math videos in A-Z Sheet (YouTube). GCD, Sieve, fast exponentiation.
Phase 3 — More Patterns		
18. Graphs (BFS & DFS)	Striver — Graph Series (YouTube, ~50 videos). Most complete free graph resource available.	Apna College — Graph playlist (YouTube). Gentler intro — do this before Striver.
19. Backtracking	Aditya Verma — Backtracking playlist (YouTube). Best for beginners, draws decision trees.	Striver — Recursion & Backtracking Series (YouTube). Pattern-wise with decision trees.
20. Tries	Striver — Tries Series (YouTube). Covers implementation and all key interview problems.	NeetCode — Tries section (neetcode.io). Concise implementation walkthrough.
Phase 4 — Advanced Algorithms		
21. Dynamic Programming	Aditya Verma — DP playlist (YouTube, watch first). Then Striver DP Series (~56 videos in strict order).	LeetCode Explore — Dynamic Programming (interactive problems with hints).
22. Greedy Algorithms	Striver — Greedy Series (YouTube). All standard greedy patterns with proofs.	GFG — Greedy Algorithms articles (geeksforgeeks.org). Good exchange argument proofs.
Phase 5 — Advanced DS		
23. Advanced Graphs	Striver — Graph Series Part 2 (YouTube, Dijkstra onwards) + CP-Algorithms Graphs articles.	cp-algorithms.com → Graphs. Best written reference for Dijkstra, Bellman-Ford, DSU, MST.
24. Segment Trees	CP-Algorithms — Segment Tree article (cp-algorithms.com). Most detailed free resource.	Striver — Segment Tree video (YouTube). Good intro before reading CP-Algorithms.
25. Advanced DP	Codeforces EDU — DP section (codeforces.com/edu). Structured lessons with problems.	Striver — Advanced DP videos (YouTube). Bitmask, tree DP, interval DP covered.